Intelligent Candidate Discovery & Ranking

Redrob Hackathon - Senior AI Engineer (Founding Team)

An explainable, CPU-only ranker that scores demonstrated work - not keywords.

Redrob Ranker - Nikhil Bhardwaj

Code: github.com/bhardwajnikhil823/redrob-ranker

Live sandbox: redrob-ranker-nikhil.streamlit.app

The problem: rank 100K candidates for a hard-to-fill role

- The JD is written to separate what it **says** from what it **means**: a perfect AI-keyword skills list behind a *Marketing Manager* title is **not** a fit; a plain-language profile that actually built a recommender **is**.
- The pool is adversarial by design: keyword stuffers, understated strong candidates, behavioral twins, and ~80 **honeypots** with subtly impossible profiles.
- Hard constraints: top-100 ranking, **CPU-only, ≤5 min, ≤16 GB, no network** - an LLM-per-candidate approach simply cannot scale here.
- Our thesis: **score demonstrated work** from career history, weigh **real availability**, and refuse to be fooled by skill-list keywords.

Architecture: a transparent scoring pipeline

1	Load 100K candidates (streamed JSONL)
2	TF-IDF semantic match + 9 structured components
3	Weighted base score in [0, 1]
4	x behavioral availability modifier (~0.5-1.1)
5	Honeypot & keyword-stuffer guards
6	Top 100 + grounded reasoning -> CSV / XLSX

Pure scikit-learn + rule logic - **no LLM, no GPU, no network**. Measured ~30 s and ~3.5 GB peak for the full 100K pool.

Scoring: nine interpretable components

Component	Weight	What it rewards
substance	0.24	Hands-on retrieval / ranking / recommendation work in the career text
semantic	0.20	TF-IDF (1-2gram) cosine similarity to a distilled JD query
title	0.16	Tiered role gate: AI/ML -> SWE -> adjacent -> non-technical
experience	0.10	5-9 year band (ideal 6-8), soft falloff outside
product	0.09	Product company vs pure IT-services / consulting career
location	0.08	Pune/Noida -> NCR -> metro -> other-India -> outside India
domain	0.07	NLP / IR (good) vs computer-vision / speech-only (negative)
evaluation	0.04	Ranking-eval experience: NDCG / MRR / MAP / A-B testing
education	0.02	Institution tier (minor)

Final score = base (sum of the nine) **x** behavioral availability modifier; honeypots are then forced out.

Beating the traps (the core differentiator)

- **Score career descriptions, not the skills array** - the exact field a stuffer inflates is excluded from the text we match against.
- **Tiered title gate** - non-technical titles are capped low; only genuine AI/ML roles get full credit (general SWE can still climb via real evidence).
- **Keyword-stuffer guard** - many claimed AI skills but no supporting career evidence -> x0.30-0.75 (**28 caught**), while genuine career-switchers are explicitly protected.
- **Honeypot detection** - logical impossibilities are forced out (x0.02):
  - role duration > its own date span ("8 yrs at a 3-yr-old company")
  - declared tenure > total experience; reversed or future dates
  - many "expert" skills with 0 months of usage

Hireability: behavioral availability modifier

- A perfect-on-paper candidate who never replies and hasn't logged in for months is, for hiring purposes, **not actually available**.
- A multiplicative modifier (~0.5-1.1x) combines five Redrob signals:
  - recruiter_response_rate · last-active recency · open-to-work
  - interview_completion_rate · profile_completeness
- Because it **multiplies** the base score, strong-but-unavailable profiles are **down-weighted, not deleted** - they can still surface if truly elite.
- This directly answers the JD's instruction to "down-weight appropriately" candidates who aren't reachable.

Explainable, grounded reasoning per candidate

- Every pick carries a 1-2 sentence justification built **only from facts in the profile**: years, title, company, strongest career evidence, top actually-used skills, location, and live signal values.
- **Honest about concerns** - out-of-band experience, outside India, services-only career, weak availability, CV/speech lean - so the tone matches the rank (checked at Stage 4).
- **No hallucination** - a skill is cited as "depth" only when career evidence supports it, otherwise neutrally as "lists".
- Result: a reviewer can audit **why** any candidate sits where they do.

Results & validation

Top-100 titles	100% technical AI/ML roles
Experience fit	82% within the 5-9 year band
Honeypots in top 100	0 (disqualification limit: 10%)
Keyword-stuffers	28 down-weighted out of contention
Compute	~30 s · ~3.5 GB peak (limits: 5 min / 16 GB)
Format & tests	passes validate_submission.py · 23 tests green

Reproducibility & engineering

- **One command**: `python rank.py --candidates candidates.jsonl --out submission.csv`
- Pinned `requirements.txt`; CPU-only; no network during ranking - reproduces inside the Stage-3 sandbox unchanged.
- **23 unit/regression tests** (incl. a honeypot false-positive regression); authentic, staged git history.
- **Live sandbox** (Streamlit) + **Dockerfile** for one-command reproduction on a small sample.
- Built to be **defended**: every rank traces to explicit, inspectable logic - not an opaque model dump.

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